

# **INTRODUCTION TO ARTIFICIAL INTELLIGENCE**



## INTRODUCTION

AI is one of the newest disciplines, formally initiated in 1956 when the name was coined. However, the study of intelligence is one of the oldest disciplines. The advent of computers made it possible for the first time for people to test models they proposed for learning, reasoning, perceiving, etc.

**Artificial Intelligence** is composed of two words **Artificial** and **Intelligence**, where **Artificial** defines "man-made," and **intelligence** defines "thinking power", hence AI means "a man-made thinking power."

## WHAT IS ARTIFICIAL INTELLIGENCE?

Artificial intelligence is **the simulation of human intelligence processes by machines, especially computer systems**. Specific applications of AI include expert systems, natural language processing, speech recognition and machine vision.

# Some Terminologies:

## **Machine Learning:**

Machine Learning is a type of technology that allows computers to learn from data and make decisions or predictions without being explicitly programmed. Its like teaching a computer to recognize patterns and improve over time based on experience.

## **Deep Learning:**

Deep Learning is a subset of machine learning that uses artificial neural networks with many layers(hence “deep”) to process and learn from large amounts of data. Each layer in the network processes information in increasingly complex ways, allowing the model to recognize patterns and make decisions

## Some Terminologies:

### Expert System:

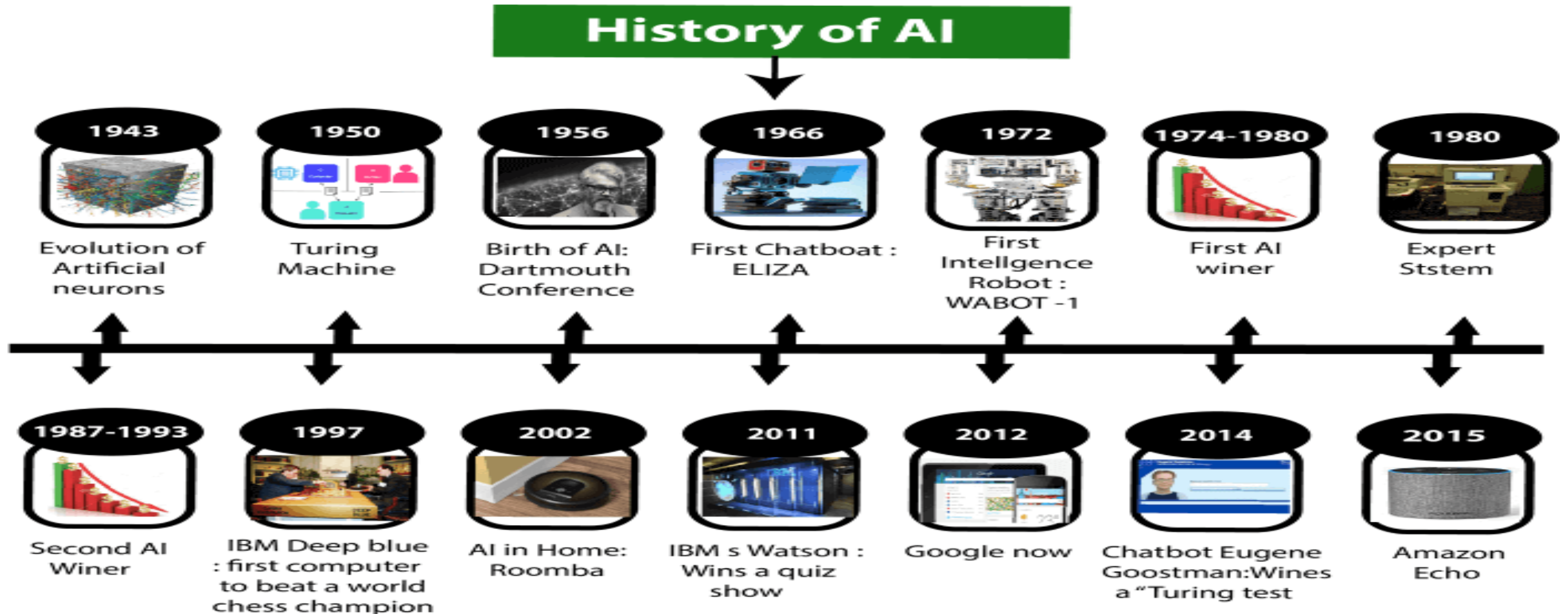
An Expert System is a type of **computer program that mimics the decision-making ability of a human expert**. It uses a set of predefined rules and knowledge to solve complex problems or give advice in a specific domain (like medical diagnosis or trouble shooting technical issues).

### Natural Language Processing:

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on **enabling computers to understand, interpret, and respond to human language in a way that is both meaningful and useful**. NLP bridges the gap between human communication and computer understanding.

# HISTORY OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence is not a new word and not a new technology for researchers. This technology is much older than you would imagine. Even there are the myths of Mechanical men in Ancient Greek and Egyptian Myths. Following are some milestones in the history of AI which defines the journey from the AI generation to till date development.



## Maturation of Artificial Intelligence (1943-1952)

- **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter Pitts in 1943. They proposed a model of artificial neurons.
- **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called Hebbian learning.
- **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes "Computing Machinery and Intelligence" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a Turing test.

## The birth of Artificial Intelligence (1952-1956)

- **Year 1955:** Allen Newell and Herbert A. Simon created the "first artificial intelligence program" which was named as "**Logic Theorist**". This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
- **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist **John McCarthy at the Dartmouth Conference**. For the first time, **AI** coined as an academic field.
- At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

## The golden years-Early enthusiasm (1956-1974)

- **Year 1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the **first chatbot in 1966**, which was named as **ELIZA**.
- **Year 1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

## The first AI winter (1974-1980)

- The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
- During AI winters, an interest of publicity on artificial intelligence was decreased.

## A boom of AI (1980-1987)

- **Year 1980:** After AI winter duration, AI came back with "**Expert System**". Expert systems were programmed that **emulate the decision-making ability of a human expert**.
- In the Year 1980, the first national conference of the American Association of Artificial Intelligence was held at Stanford University.

## The second AI winter (1987-1993)

- The duration between the years 1987 to 1993 was the second AI Winter duration.
- Again Investors and government stopped in funding for AI research as due to high cost but not efficient result. The expert system such as XCON was very cost effective.

## The emergence of intelligent agents (1993-2011)

- Year 1997: In the year 1997, **IBM Deep Blue beats world chess champion**, Gary Kasparov, and became the first computer to beat a world chess champion.
- **Year 2002: for the first time**, AI entered the home in the form of **Roomba**, a vacuum cleaner.
- Year 2006: AI came in the Business world till the year 2006. Companies like Facebook, Twitter, and Netflix also started using AI.

## Deep learning, big data and artificial general intelligence (2011-present)

- **Year 2011:** In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. **Watson had proved that it could understand natural language and can solve tricky questions quickly.**
- **Year 2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- **Year 2014:** In the year 2014, **Chatbot "Eugene Goost man"** won a competition in the infamous "Turing test."
- **Year 2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.
- Google has demonstrated an AI program "**Duplex**" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.

# FOUNDATIONS OF AI

Like any history, this one is forced to concentrate on a small number of people, events, and ideas and to ignore others that also were important. We organize the history around a series of questions. We certainly would not wish to give the impression that these questions are the only ones the disciplines address or that the disciplines have all been working toward AI as their ultimate fruition.

## 1. PHILOSOPHY

- ✓ Can formal rules be used to draw valid conclusions?
- ✓ How does the mind arise from a physical brain?
- ✓ Where does knowledge come from?
- ✓ How does knowledge lead to action?

**Aristotle (384–322 B.C.)**, was the first to formulate a precise set of laws governing the rational part of the mind. He developed an informal system of syllogisms for proper reasoning, which in principle allowed one to generate conclusions mechanically, given initial premises.

Much later, Ramon Lull (d. 1315) had the idea that useful reasoning could actually be carried out by a mechanical artifact. Thomas Hobbes (1588–1679) proposed that reasoning was like numerical computation that “we add and subtract in our silent thoughts.”

## 2 MATHEMATICS

- ✓ What are the formal rules to draw valid conclusions?
- ✓ What can be computed?
- ✓ How do we reason with uncertain information?

Philosophers staked out most of the important ideas of AI, but to move to a formal science requires a level of mathematical formalism in three main areas: **computation, logic and probability**.

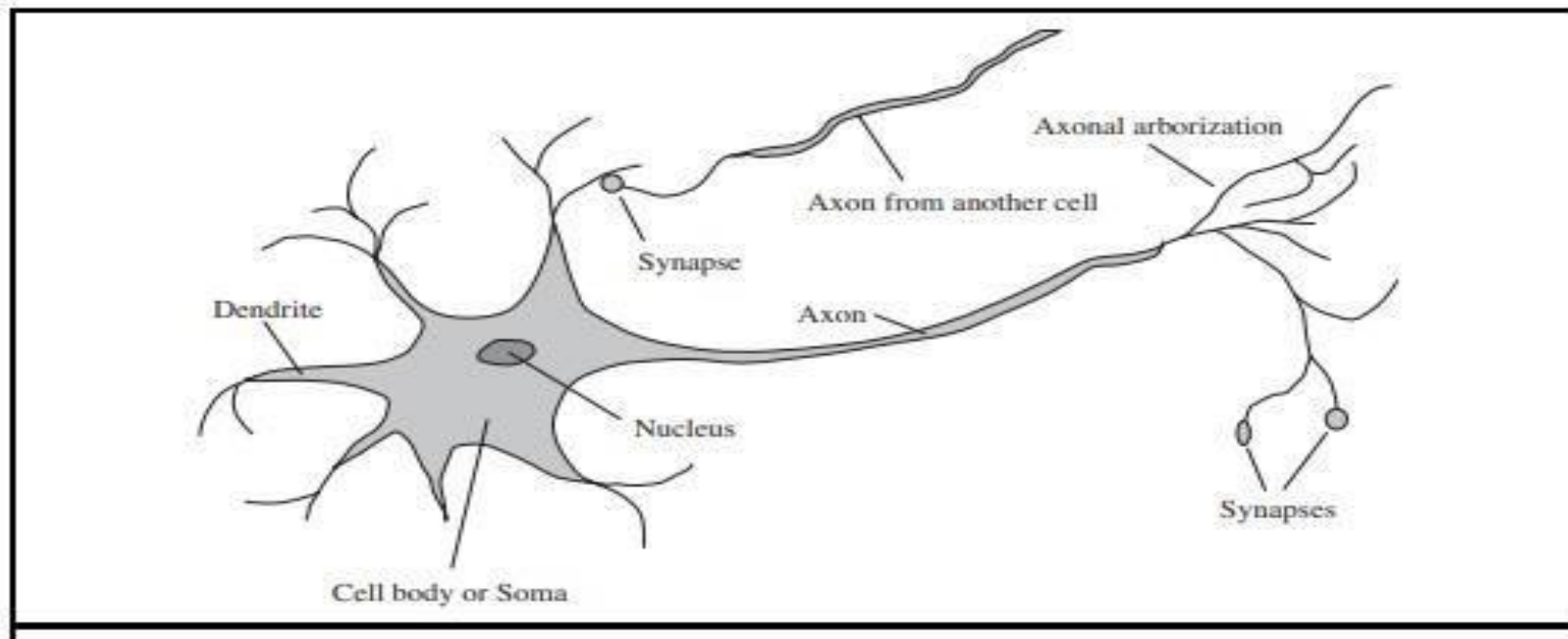
Mathematicians have proved that there exists an algorithm to prove **any true statement in first-order logic**. However, if one adds the principle of induction required to capture the semantics of the natural numbers, then this is no longer the case. Specifically, the **incompleteness theorem** showed that in any language expressive enough to describe the properties of the natural numbers, there are true statements that are undecidable: their truth cannot be established by any algorithm.

### 3 ECONOMICS

- ✓ How should we make decisions so as to maximize payoff?
- ✓ How should we do this when others may not go along?
- ✓ How should we do this when the payoff may be far in the future?

### 4 NEUROSCIENCE

- ✓ How do brains process information?
- Neuroscience is the study of the nervous system, particularly the brain. Although the exact way in which the brain enables thought is one of the great mysteries of science, the fact that it does enable thought has been appreciated for thousands of years because of the evidence that strong blows to the head can lead to mental incapacitation.
- It has also long been known that human brains are somehow different; Aristotle wrote, “Of all the animals, man has the largest brain in proportion to his size.” Still, it was not until the middle of the 18th century that the brain was widely recognized as the seat of consciousness. Before then, candidate locations included the heart and the spleen.



The parts of a nerve cell or neuron. Each neuron consists of a cell body, or soma, that contains a cell nucleus. Branching out from the cell body are a number of fibers called dendrites and a single long fiber called the axon. The axon stretches out for a long distance, much longer than the scale in this diagram indicates.

|                     | Supercomputer                             | Personal Computer                         | Human Brain                             |
|---------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------|
| Computational units | $10^4$ CPUs, $10^{12}$ transistors        | 4 CPUs, $10^9$ transistors                | $10^{11}$ neurons                       |
| Storage units       | $10^{14}$ bits RAM<br>$10^{15}$ bits disk | $10^{11}$ bits RAM<br>$10^{13}$ bits disk | $10^{11}$ neurons<br>$10^{14}$ synapses |
| Cycle time          | $10^{-9}$ sec                             | $10^{-9}$ sec                             | $10^{-3}$ sec                           |
| Operations/sec      | $10^{15}$                                 | $10^{10}$                                 | $10^{17}$                               |
| Memory updates/sec  | $10^{14}$                                 | $10^{10}$                                 | $10^{14}$                               |

## 5 PSYCHOLOGY

- ✓ How do humans and animals think and act?

The principle characteristic of **cognitive psychology** is that the brain processes and processes information. The claim is that beliefs, goals, and reasoning steps can be useful components of a theory of human behaviour. The knowledge-based agent has three key steps:

1. Stimulus is translated into an internal representation
2. The representation is manipulated by cognitive processes to derive new internal representations

These are translated into actions

## 6 COMPUTER ENGINEERING

- ✓ How can we build an efficient computer?

For artificial intelligence to succeed, we need two things: intelligence and an artifact. The computer has been the artifact of choice.

## 7 CONTROL THEORY AND CYBERNETICS

- ✓ How can artifacts operate under their own control?

Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize an objective function over time. This roughly OBJECTIVE FUNCTION matches our view of AI: **designing systems that behave optimally. Why, then, are AI and control theory two different fields**, despite the close connections among their founders? The answer lies in the close coupling between the mathematical techniques that were familiar to the participants and the corresponding sets of problems that were encompassed in each world view.

## 8. LINGUISTICS

- ✓ Having a theory of how humans successfully process natural language is an AI-complete problem - if we could solve this problem then we would have created a model of intelligence.
- ✓ Much of the early work in knowledge representation was done in support of programs that attempted natural language understanding.

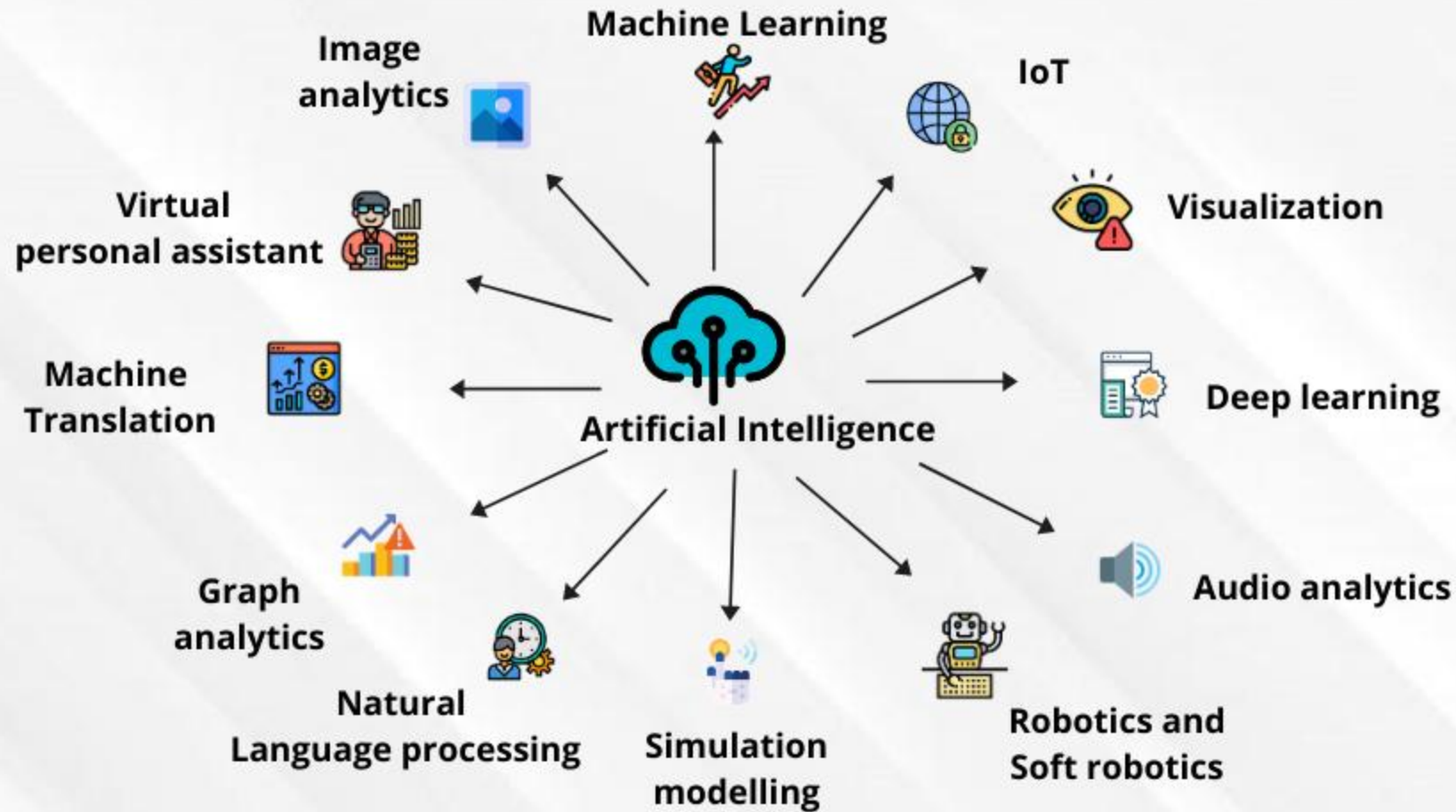
# Evaluation of AI

- Evaluation of AI systems is an important task that involves measuring their performance against some criteria.
- These criteria can be of many different types, depending on the application and the intended users.
- Some common criteria include **accuracy, speed, reliability, scalability, usability, maintainability, and cost.**
- The key is to **choose the criteria** that are most relevant to the users and that **can provide meaningful insights into the strengths, weaknesses, and trade-offs** of the AI system

# Scope of AI

- The scope of AI is vast and includes many subfields such as **knowledge representation, search, planning, perception, natural language processing, learning, robotics**, and many more.
- Some of the most **active areas of research** in AI today include **machine learning, deep learning, reinforcement learning, neural networks, computer vision, natural language understanding, and expert systems**.
- AI is also being used in many **application areas** such as autonomous vehicles, healthcare, finance, education, entertainment, and security.
- However, AI is still an active and evolving field and there is much that is still not fully understood or solved.

# Scope of Artificial Intelligence



CYBER CHASSE

# Overview of AI problems

- An overview of AI problems includes several types of tasks that AI systems may be designed to address, such as **classification, regression, clustering, anomaly detection, association rule mining, sequence prediction, natural language understanding, and multi-agent decision making.**
- In general, these tasks can be formulated as **optimization problems**, where the **objective is to find the best solution** according to some criteria, such as accuracy, speed, or simplicity.

# Overview of AI problems

- Different methods can be used to solve these problems, including **rule-based system, decision trees, neural networks, support vector machines, and evolutionary algorithms.**
- The choice of method depends on the nature of the problem and the available data, among other factors.

# Components of AI

- The main parts of AI that work together to make the system intelligent are as follows:

## 1. Learning in AI

Learning in Artificial Intelligence (AI) refers to the process by which an AI system improves its performance over time by analyzing data, identifying patterns and making decisions based on past experience. It helps them to adapt and change without being explicitly programmed for every task.

There are different types of learning such as Supervised Learning, Unsupervised Learning, Reinforcement Learning, Semi-Supervised Learning, Self-Supervised Learning, Deep Learning .

# Components of AI(Cont..)

## 2. Reasoning in AI

Reasoning helps AI to make logical decisions, solve problems and infer conclusions based on given information much like human reasoning. It allows AI to use information and rules to come to conclusions just like we humans do.

There are different types of Reasoning such as Deductive Reasoning, Inductive Reasoning, Probabilistic Reasoning etc.

## 3. Ethics and Bias in AI

Systems learn from data but the data they use can have biases. It's important to mention that AI systems need to be carefully designed and monitored to avoid unfair biases or making unethical decisions.

**Ethical Considerations:** Systems must be designed responsibly to ensure they do not harm individuals or groups and their decisions should be transparent.

**Bias in AI:** If the training data is biased AI might make biased predictions (e.g racial bias in facial recognition software).

# Components of AI(Cont...)

## 4. Problem Solving in AI

helps it in identifying complex problems and finding ways to solve them efficiently. System use algorithms designed to solve problems across various domains such as Search Algorithms, Optimization Techniques, Simulation and Modeling etc.

## 5. Perception in AI

Perception allows systems to understand their surroundings by processing sensory inputs such as images, sounds and even touch. This is important for AI to interact with the real world.

Main Aspects of AI include Computer Vision which helps AI to interpret and understand visual input from the surrounding environment, eg. Facial recognition, autonomous vehicle and surveillance systems.

Speech Recognition: Allows AI to covert spoken language into text. Generally used in virtual assistant technologies, automated customer support and real time communication devices.

# Components of AI(Cont..)

## 6. Data in AI

Data is considered as the **fuel** that helps AI systems. Accuracy, effectiveness and fairness of AI models are influenced by the quality of the data used to train them. Without good data even the most complex AI algorithms can produce poor or biased results.

Types of data used in AI can be structured or unstructured.

## 7. Language Processing

Language processing which is the part of natural language processing (NLP) helps machines to read, understand and produce human language. It involves tasks from basic text analysis to complex conversations.

core NLP functions include Syntax Analysis, Semantic Analysis, Dialogue management etc.

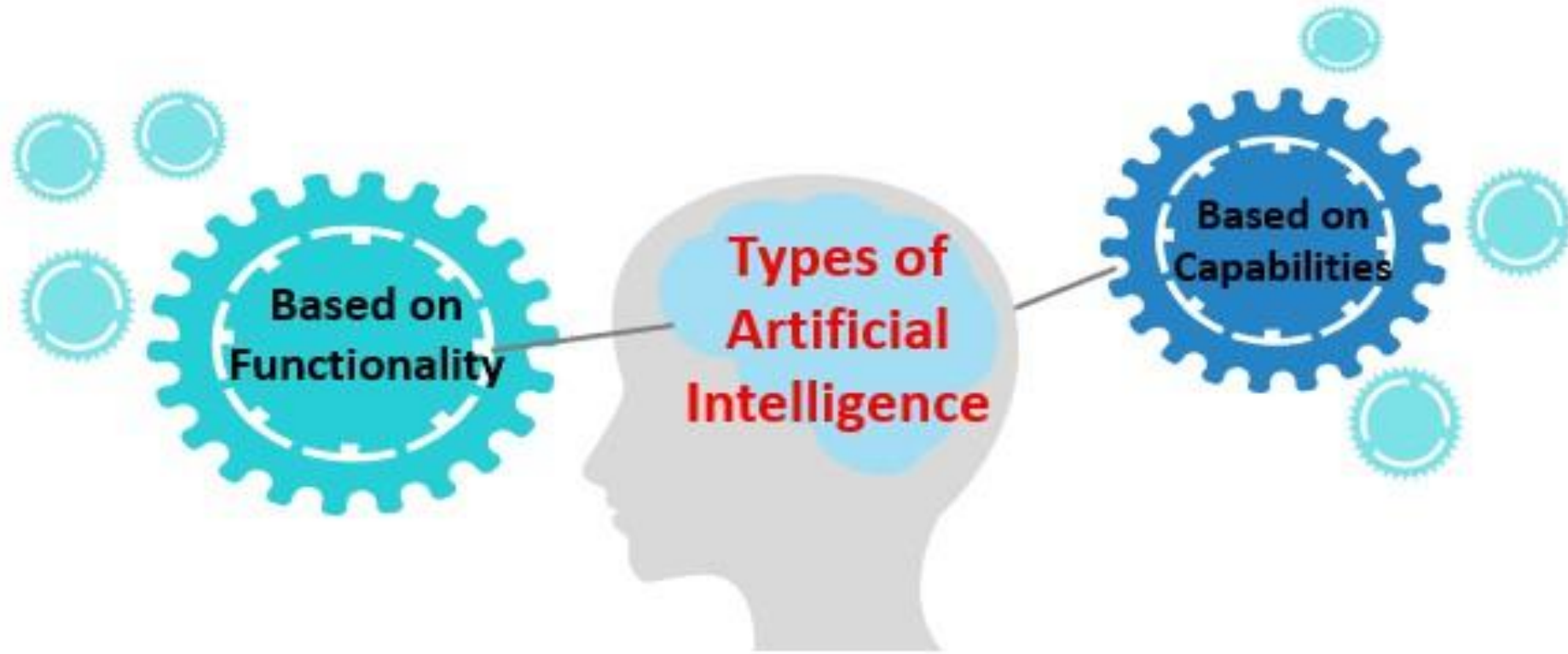
# Types of Artificial Intelligence



# TYPES OF ARTIFICIAL INTELLIGENCE

The main aim of Artificial Intelligence is to enable machines to perform a human-like function

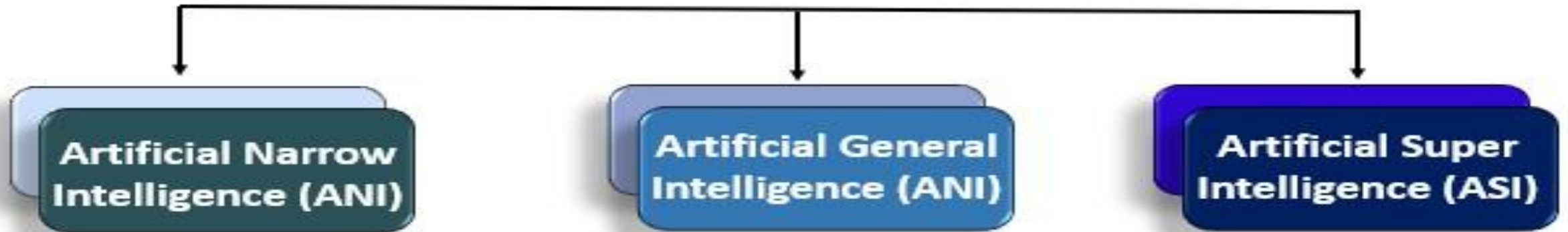
Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are **based on capabilities** and **based on functionality** of AI.



## Type 1 - Based on Functionality



## Type 2 - Based on Capabilities



# BASED ON FUNCTIONALITY

## 1. REACTIVE MACHINES

- Purely reactive machines are the most **basic types** of Artificial Intelligence.
- Such AI systems **do not store memories** or **past experiences** for **future actions**.
- These machines **only focus** on **current scenarios** and **react on it** as per possible **best action**.
- **IBM's Deep Blue system** is an example of reactive machines.
- **Google's AlphaGo** is also an example of reactive machines.

## 2. LIMITED MEMORY

- Limited memory machines can store **past experiences** or **some data** for **a short period** of time.
- These machines can use stored data for a **limited time period only**.
- **Self-driving cars are one of the best examples** of Limited Memory systems. These cars can store **recent speed of nearby cars**, the **distance of other cars**, **speed limit**, and other information to navigate the road.

### 3. THEORY OF MIND

- Theory of Mind AI should understand the **human emotions, people, beliefs**, and be able to **interact socially like humans**.
- This **type of AI machines are still not developed**, but researchers are making lots of efforts and improvement for developing such AI machines.

### 4. SELF-AWARENESS

- Self-awareness AI is the **future** of **Artificial Intelligence**. These machines will be **super intelligent**, and will have their own **consciousness, sentiments**, and **self-awareness**.
- These machines will be **smarter than human mind**.
- Self-Awareness **AI does not exist in reality still** and it is a **hypothetical concept**.

## BASED ON CAPABILITIES

### 1. WEAK AI OR NARROW AI:

- Narrow AI is a type of AI which is able to perform a **dedicated task** with intelligence. **The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.**
- Narrow AI cannot **perform beyond its** field or **limitations**, as it is only **trained for one specific task**. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.
- **Apple Siri** is a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
- **IBM's Watson supercomputer** also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
- **Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.**

## 2. GENERAL AI:

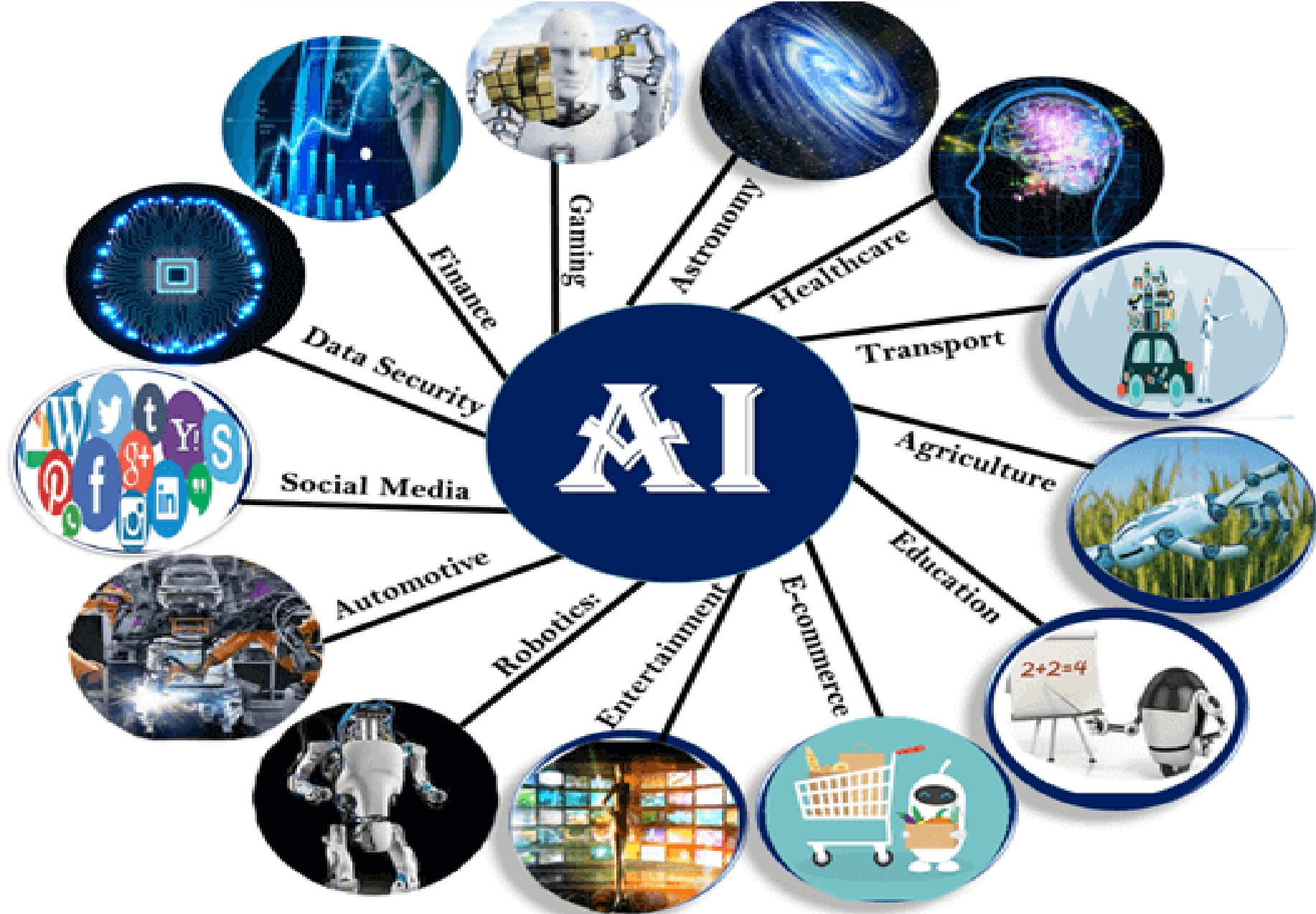
- General AI is a type of intelligence which could perform any **intellectual task with efficiency like a human.**
- The idea behind the general AI to make such a system which could be **smarter** and **think like a human by its own.**
- Currently, there is **no such system exist which could come under general AI** and can perform any task **as perfect as a human.**
- The worldwide researchers are now focused on developing machines with General AI.
- As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.

### 3. SUPER AI:

- Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an outcome of general AI.
- Some key characteristics of strong AI capability include the **ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.**
- Super AI is still a **hypothetical concept** of Artificial Intelligence. Development of such systems in real is still world changing task.



# APPLICATION OF AI



## **1. AI in Astronomy**

Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

## **2. AI in Healthcare**

In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.

Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

## **3. AI in Gaming**

AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

#### **4. AI in Finance**

AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

#### **5. AI in Data Security**

The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

#### **6. AI in Social Media**

Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

## 7. AI in Travel & Transport

AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

## 8. AI in Automotive Industry

Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as **Tesla has introduced TeslaBot, an intelligent virtual assistant.**

Various Industries are currently working for **developing self-driven cars** which can make your journey more safe and secure.

## 9. AI in Robotics:

Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.

Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as **Erica and Sophia** has been developed which can talk and behave like humans.

## 10. AI in Entertainment

We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

## **11. AI in Agriculture**

Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers

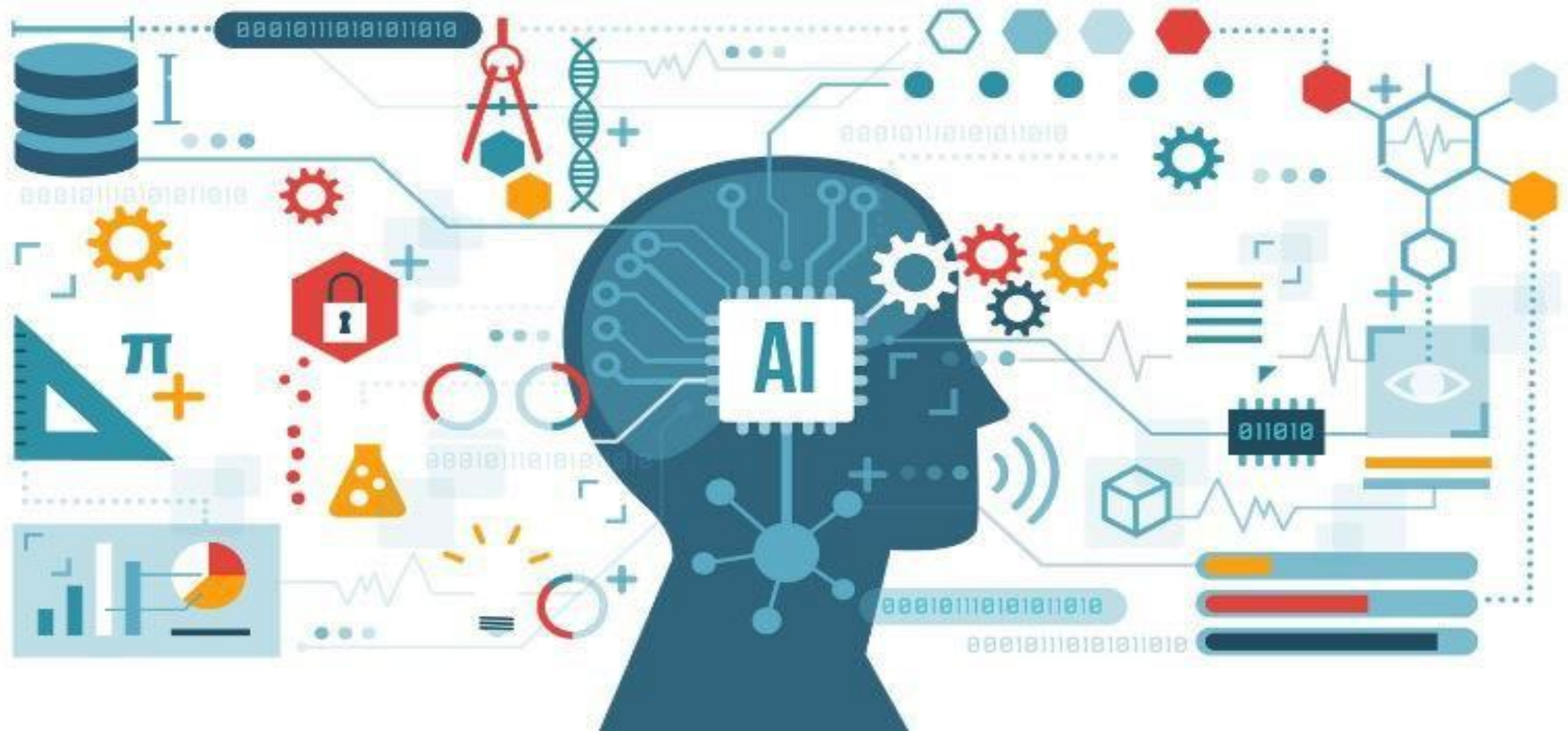
## **12. AI in E-commerce**

AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

## **13. AI in education:**

AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.

# BENEFITS OF ARTIFICIAL INTELLIGENCE



# **BENEFITS OF ARTIFICIAL INTELLIGENCE**

- 1. INCREASED EFFICIENCY**
- 2. REDUCING HUMAN RISK**
- 3. AID IN DECISION MAKING**
- 4. AVAILABILITY**

## **1. INCREASED EFFICIENCY**

Machines are highly efficient in contrast to the human workforce. This is because they can complete monotonous tasks effectively without taking a break and irrespective of the working conditions.

Take the example of Power BI. Earlier, time was spent in **sifting, isolating, and utilizing the information** in **MS Excel**, **Power BI** utilizes AI to do the entirety of this (and significantly more) autonomously.

This expands the effectiveness of the framework and spares significant time for the investigator to utilize it somewhere else.

## 2. REDUCING HUMAN RISK

A lot of places wherein it is **hazardous** for a person to be truly present.

Man-made brainpower can be useful in such cases since it would empower the machines to have the option to settle on choices continuously. A case of this is a remote ocean boring at the sea bed where there is a high danger of human life.

## 3. AID IN DECISION MAKING

This is the domain where AI has successfully paid dividends. AI has the capacity to analyze a great deal of information in a brief time frame. This is helping people improve decision making. **Take the case of an investor who needs to conclude whether to give out a credit or not.**

This can be highly effective in verticals like **portfolio management**. This would help in a more astute and quicker dynamic for the investor.

## 4. AVAILABILITY

Machines, in contrast to people, don't have to take rests. They can work nonstop.

Now, we can depend on machines for keeping up required manufacturing units running with their judgment which would prompt 24×7 creation units and complete mechanization.



***Improve efficiency***



***Save costs***



***Avoid problems***



***Anticipate future scenarios***



***Reduce human error***



***Automate processes***



# **AGENTS** **AND** **ENVIRONMENTS**

# AGENTS AND ENVIRONMENTS

- **Agents** in Artificial Intelligence are the **associated concepts** that the **AI technologies** work upon.
- The **AI software or AI-enabled devices** with **sensors** generally captures the information **from** the **environment setup** and **process** the **data for further actions**.
- There are mainly two ways the agents interact with the environment, such as **perception** and **action**.
- The **perception** is only **passive** for capturing the **information** without **changing** the **actual environment**, whereas **action** is the **active** form of **interaction** by **changing** the **actual environment**.
- AI technologies such as **virtual assistance catboats**, AI-enabled devices to work based on the **previous persecution data processing and learning for the actions**.

## WHAT IS AN AGENT?

An Agent is anything that takes actions according to the information that it gains from the environment.

- **HUMAN-AGENT:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **ROBOTIC AGENT:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **SOFTWARE AGENT:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

## EXAMPLE FOR AGENT AND ENVIRONMENTS?

Agents

Environments

Robot → Room

Chatbot → Chatting

Vehicle → Road

Program → Data & Rules

Machine → Working Field

# HOW DOES THE AGENT INTERACT WITH THE ENVIRONMENT?

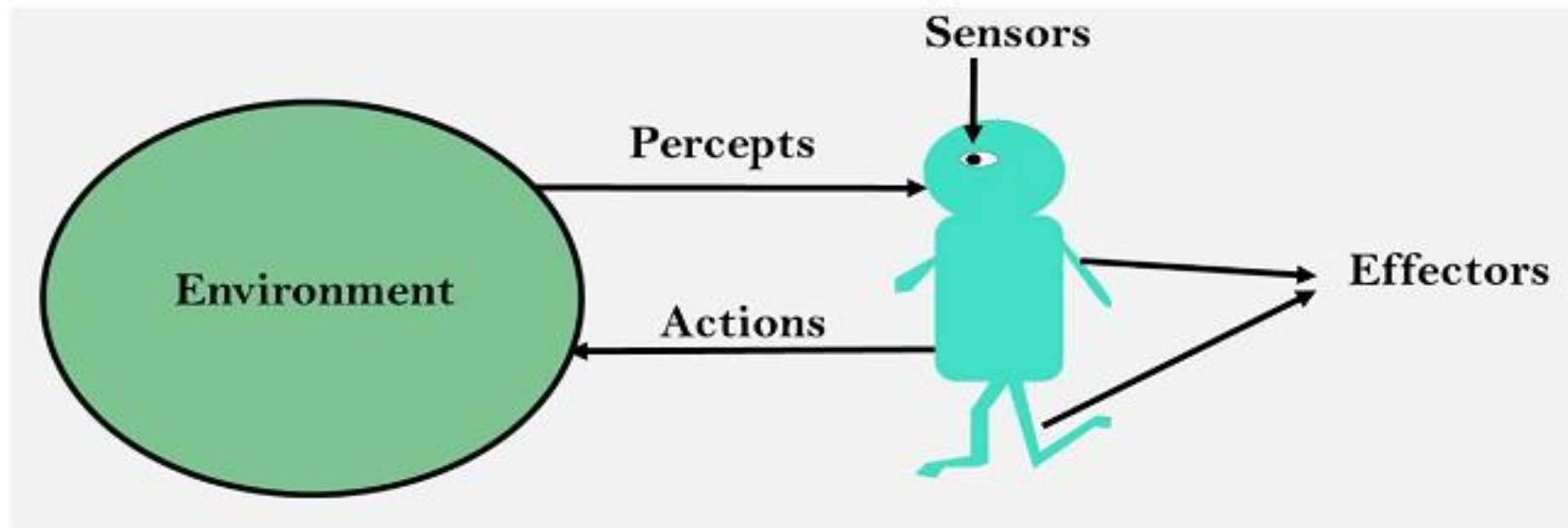
The agents interact with the environment in two ways:

## 1. PERCEPTION

**Perception** is a **passive interaction**, where the **agent gains information** about the **environment without changing the environment**. The **sensors** of the **robot help** it to **gain information** about the **surroundings** without **affecting** the **surrounding**. Hence, gaining information through sensors is called perception.

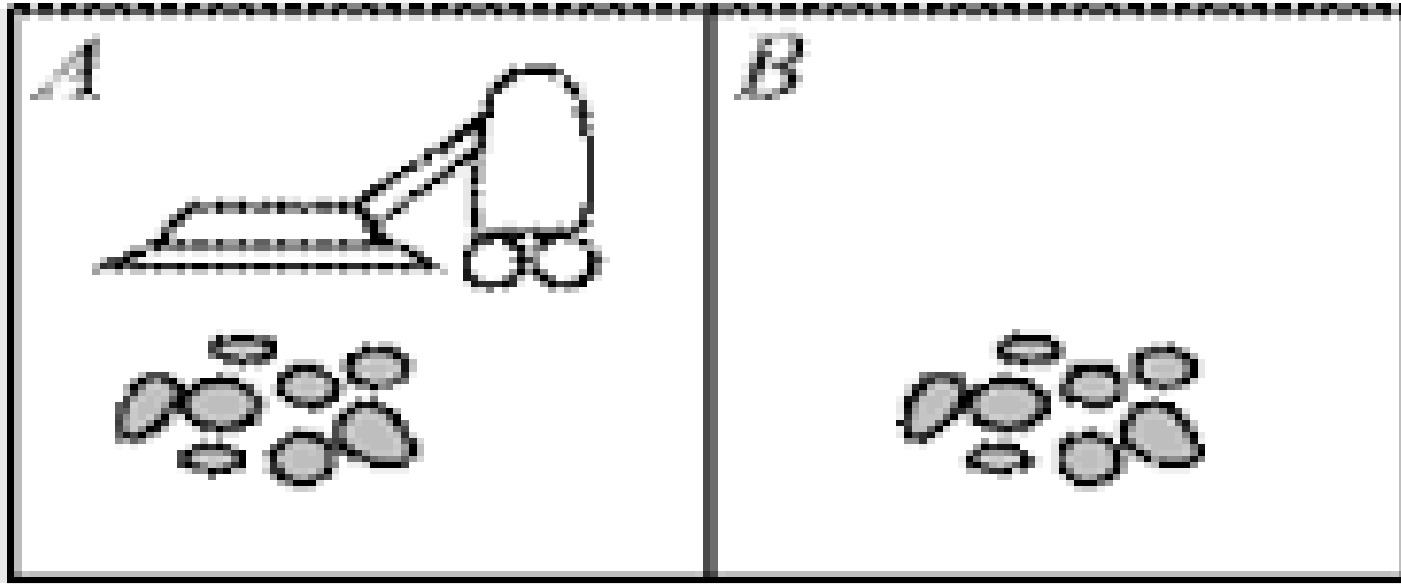
## 2. ACTION

**Action** is an **active interaction** where the **environment is changed**. When the **robot moves** an **obstacle** using its **arm**, it is called an **action** as the **environment is changed**. The arm of the robot is called an **“Effector”** as it performs the action.



- **SENSOR:** Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.
- **ACTUATORS:** Actuators (like electrical, hydraulic, or pneumatic power) are the component of machines that convert energy into motion. The actuators are only **responsible for moving and controlling a system**. An actuator can be an **electric motor, gears, rails**, etc.
- **EFFECTORS:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

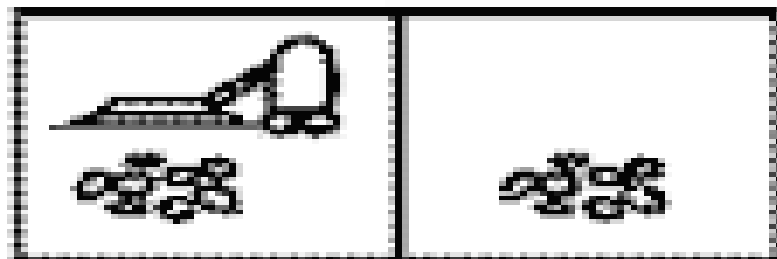
## CONSIDER A VACUUM CLEANER WORLD



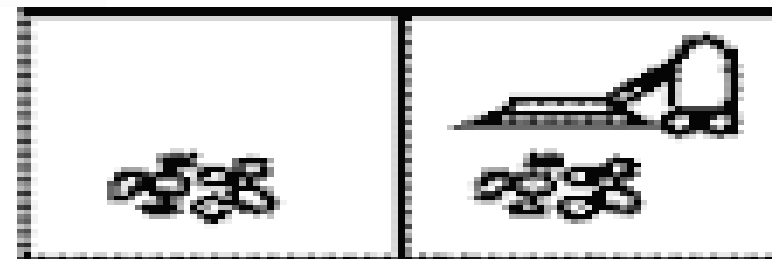
Imagine that our **intelligent agent** is a robot vacuum cleaner.

Let's suppose that the world has just two rooms. The robot can be in either room and there can be dirt in zero, one, or two rooms.

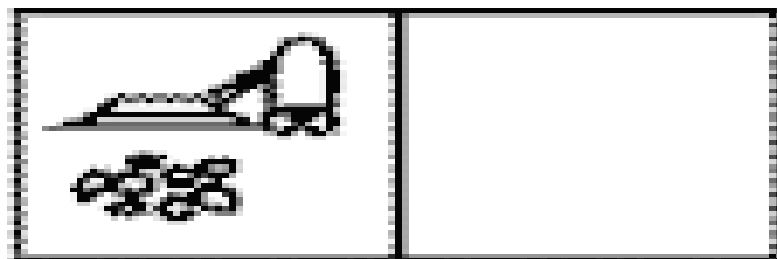
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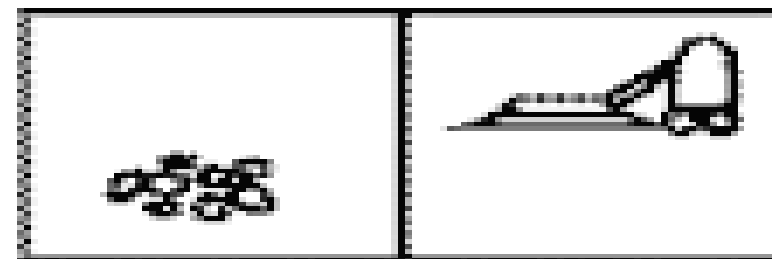
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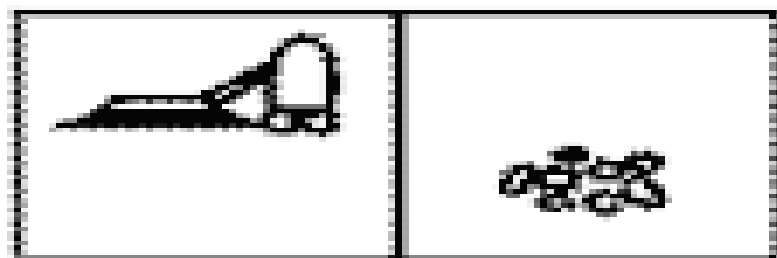
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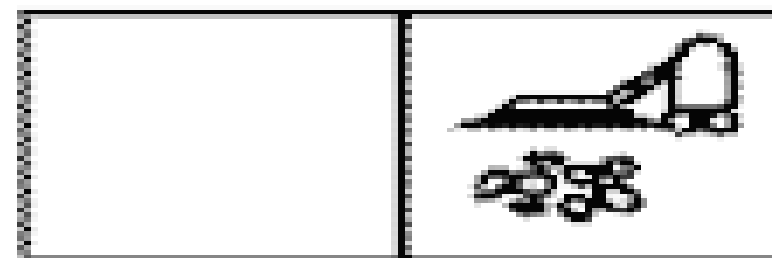
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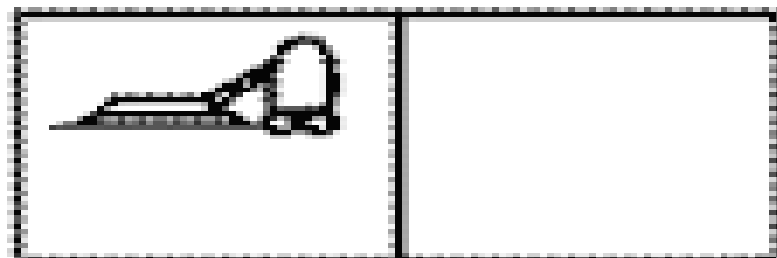
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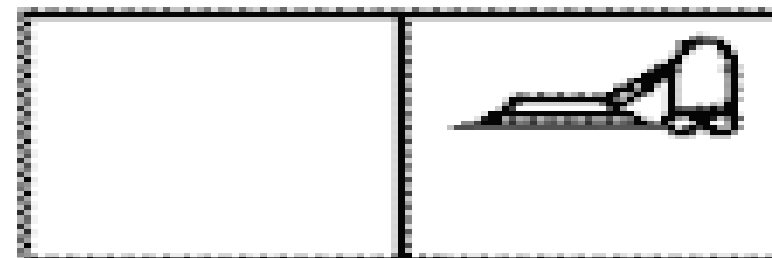
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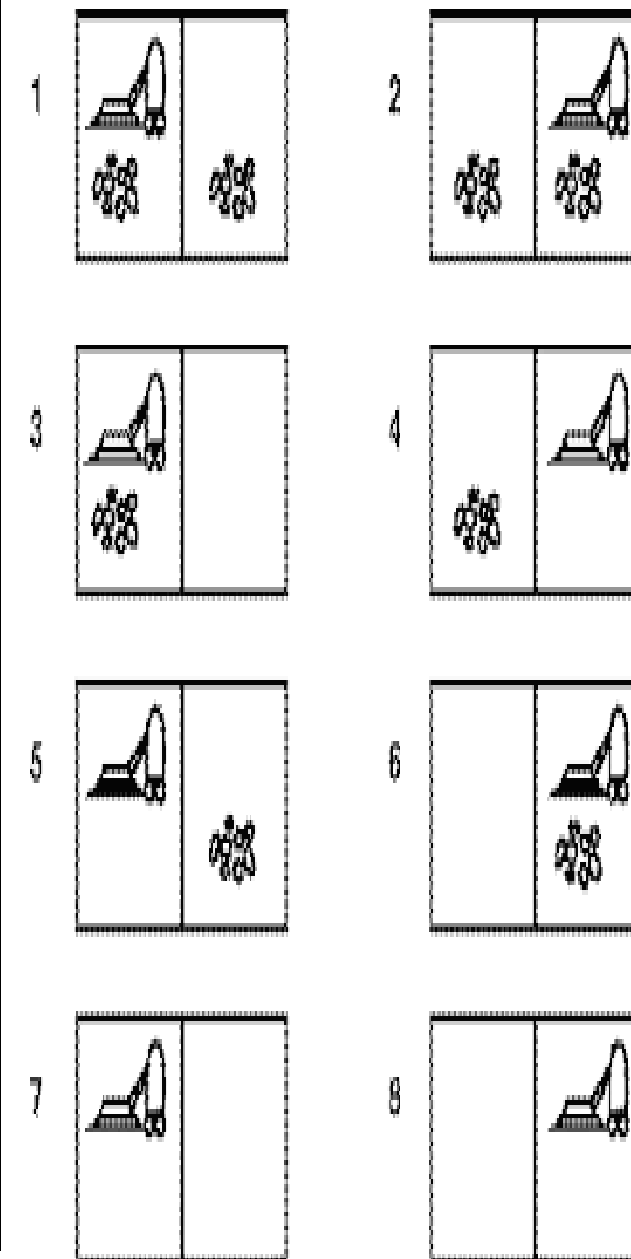
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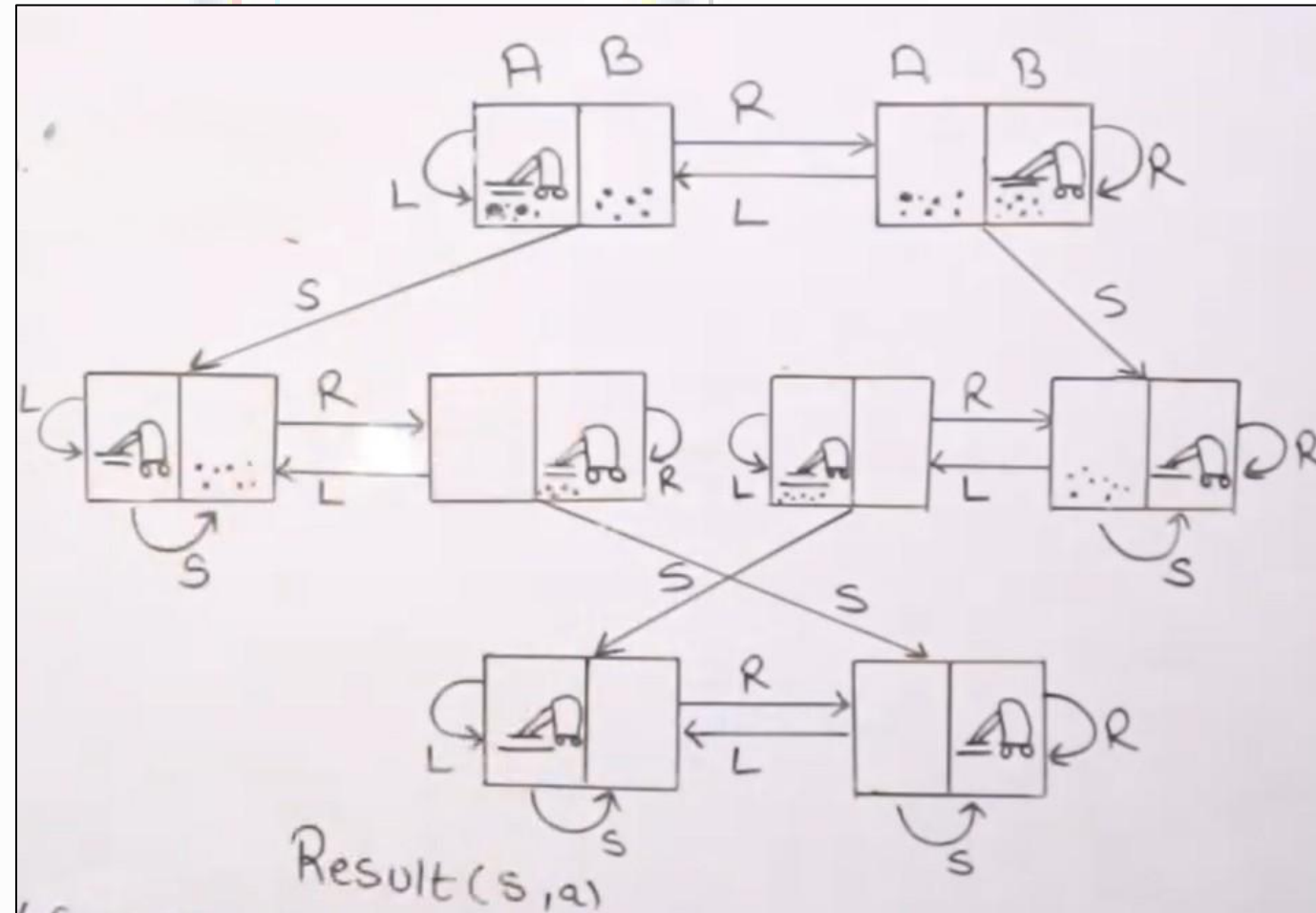
| Percept sequence                                                           | Action       |
|----------------------------------------------------------------------------|--------------|
| $[A, \textit{Clean}]$                                                      | <i>Right</i> |
| $[A, \textit{Dirty}]$                                                      | <i>Suck</i>  |
| $[B, \textit{Clean}]$                                                      | <i>Left</i>  |
| $[B, \textit{Dirty}]$                                                      | <i>Suck</i>  |
| $[A, \textit{Clean}], [A, \textit{Clean}]$                                 | <i>Right</i> |
| $[A, \textit{Clean}], [A, \textit{Dirty}]$                                 | <i>Suck</i>  |
| $\vdots$                                                                   | $\vdots$     |
| $[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Clean}]$            | <i>Right</i> |
| $[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Dirty}]$            | <i>Suck</i>  |
| $\vdots$                                                                   | $\vdots$     |
| Partial tabulation of a simple agent function for the vacuum-cleaner world |              |



Goal formulation: intuitively, we want all the dirt cleaned up. Formally, the goal is {State 7, state 8}.

Problem formulation (Actions): Left, Right, Suck, NoOp

## STATE SPACE GRAPH:



```
function REFLEX-VACUUM-AGENT( [location,status]) returns  
action
```

```
    if status = Dirty then return Suck  
    else if location = A then return Right  
    else if location = B then return Left
```

## MEASURING PERFORMANCE

With any intelligent agent, we want it to find a (good) solution and not spend forever doing it.

The interesting quantities are, therefore,

**THE SEARCH COST**--how long the agent takes to come up with the solution to the problem,  
and

**THE PATH COST**--how expensive the actions in the solution are.

The total cost of the solution is the sum of the above two quantities.

# HOW SHOULD THE AGENTS ACT IN ARTIFICIAL INTELLIGENCE?

- A rational agent does the **right thing**. The right action is the one that causes the agent to be the most successful.
- An omniscient agent knows what impact the action will have and can act accordingly, but it is not possible in reality.
- The degree of success which is defined by the performance measure
- The percept sequence which is the entire sequence of perceptions by the agent until the present moment

| Agent Type | Percepts                                    | Action                                                       | Goals                               | Environment                       |
|------------|---------------------------------------------|--------------------------------------------------------------|-------------------------------------|-----------------------------------|
| Car Driver | Speedometer, GPS,<br>Microphone,<br>Cameras | Steering control,<br>accelerate, brake,<br>talk to passenger | Safe, legal,<br>comfortable journey | Road, Traffic,<br>Pedestrian etc. |

# GOOD BEHAVIOUR: THE CONCEPT OF RATIONALITY

## INTELLIGENT AGENTS:

An **intelligent agent** is an autonomous entity which act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

### Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

## RATIONAL AGENT:

- A rational agent is an agent which has **clear preference**, models uncertainty, and acts in a way to **maximize** its **performance measure with all possible actions**.
- A rational agent is said to perform the **right things**. AI is about creating **rational agents** to use for **game theory** and **decision theory** for **various real- world scenarios**.
- For an AI agent, the **rational action is most important** because in AI reinforcement learning algorithm, **for each best possible action**, agent **gets the positive reward** and for each wrong action, an **agent gets a negative reward**.

**NOTE:** Rational agents in AI are very similar to intelligent agents.

## RATIONALITY:

The rationality of an agent is measured by its performance measure. Rationality can be judged on the basis of following points:

- Performance measure which defines the success criterion.
- Agent prior knowledge of its environment.
- Best possible actions that an agent can perform.
- The actions that the agent can perform.
- The agent's percept sequence to date.

**NOTE:** Rationality differs from Omniscience because an Omniscient agent knows the actual outcome of its action and act accordingly, which is not possible in reality.

# TASK ENVIRONMENTS

To design a rational agent we need to specify a task environment

- A problem specification for which the agent is a solution

## PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

Here performance measure is the objective for the success of an agent's behaviour.

# PEAS: SPECIFYING AN AUTOMATED TAXI DRIVER

**Performance measure: ?**

**Environment: ?**

**Actuators: ?**

**Sensors: ?**

**Performance measure:**

- safe, fast, legal, comfortable, maximize profits

**Environment:**

- roads, other traffic, pedestrians, customers

**Actuators:**

- steering, accelerator, brake, signal, horn

**Sensors:**

- cameras, sonar, speedometer, GPS



# PEAS: MEDICAL DIAGNOSIS SYSTEM

- **PERFORMANCE MEASURE:** Healthy patient, minimize costs, lawsuits
- **ENVIRONMENT:** Patient, hospital, staff
- **ACTUATORS:** Screen display (form including: questions, tests, diagnoses, treatments, referrals)
- **SENSORS:** Keyboard (entry of symptoms, findings, patient's answers)

## Example of Agents with their PEAS representation

| Agent Type                      | Performance Measure                 | Environment                      | Actuators                                                     | Sensors                                                 |
|---------------------------------|-------------------------------------|----------------------------------|---------------------------------------------------------------|---------------------------------------------------------|
| Medical diagnosis system        | Healthy patient, reduced costs      | Patient, hospital, staff         | Display of questions, tests, diagnoses, treatments, referrals | Keyboard entry of symptoms, findings, patient's answers |
| Satellite image analysis system | Correct image categorization        | Downlink from orbiting satellite | Display of scene categorization                               | Color pixel arrays                                      |
| Part-picking robot              | Percentage of parts in correct bins | Conveyor belt with parts; bins   | Jointed arm and hand                                          | Camera, joint angle sensors                             |
| Refinery controller             | Purity, yield, safety               | Refinery, operators              | Valves, pumps, heaters, displays                              | Temperature, pressure, chemical sensors                 |
| Interactive English tutor       | Student's score on test             | Set of students, testing agency  | Display of exercises, suggestions, corrections                | Keyboard entry                                          |

Examples of agent types and their PEAS descriptions.



# **THE NATURE OF ENVIRONMENTS**

# THE NATURE OF ENVIRONMENTS

- An environment in artificial intelligence is **the surrounding of the agent**. **The agent takes input from the environment through sensors** and **delivers the output to the environment through actuators**.
- An environment is **everything** in the world **which surrounds the agent**, but it is not a part of an **agent itself**. An environment can be described as a situation in which an agent is present.
- The environment is where agent lives, operate and provide the agent with something to sense and act upon it. **An environment is mostly said to be non-feministic.**

# FEATURES OF ENVIRONMENT

As per Russell and Norvig, an environment can have various features from the point of view of an agent:

1. Fully observable vs Partially Observable
2. Static vs Dynamic
3. Discrete vs Continuous
4. Deterministic vs Stochastic
5. Single-agent vs Multi-agent
6. Episodic vs sequential
7. Known vs Unknown
8. Accessible vs Inaccessible

## 1. FULLY OBSERVABLE VS PARTIALLY OBSERVABLE:

- ✓ If an agent sensor can sense or access the complete state of an environment at each point of time then it is a **fully observable** environment, else it is **partially observable**.
- ✓ A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world. Eg. Chess is fully observable, Card game, car driving is partially observable
- ✓ An agent with no sensors in all environments then such an environment is called as **unobservable**.

## 2. DETERMINISTIC VS STOCHASTIC:

- ✓ If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.
- ✓ A stochastic environment is random in nature and cannot be determined completely by an agent.
- ✓ In a deterministic, fully observable environment, agent does not need to worry about uncertainty.
- ✓ E.g. Tic Tac Toe is Deterministic but Ludo is Stochastic

### 3. EPISODIC VS SEQUENTIAL:

- ✓ In an episodic environment, there is a series of one-shot actions, and only the current percept is required for the action.
- ✓ However, in Sequential environment, an agent requires memory of past actions to determine the next best actions. Part Picking Robot is Episodic but Chess is sequential

### 4. SINGLE-AGENT VS MULTI-AGENT

- ✓ If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- ✓ However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- ✓ The agent design problems in the multi-agent environment are different from single agent environment.

## 5. STATIC VS DYNAMIC:

- ✓ If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
- ✓ Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
- ✓ However for dynamic environment, agents need to keep looking at the world at each action.
- ✓ Taxi driving is an example of a dynamic environment whereas Crossword puzzles, are an example of a static environment.

## 6. DISCRETE VS CONTINUOUS:

- ✓ If in an environment there are a finite number of percepts and actions that can be performed within it, then such an environment is called a discrete environment else it is called continuous environment.
- ✓ A chess game comes under discrete environment as there is a finite number of moves that can be performed.
- ✓ A self-driving car is an example of a continuous environment.

## 7. ACCESSIBLE VS INACCESSIBLE

- ✓ If an agent can obtain complete and accurate information about the state's environment, then such an environment is called an Accessible environment else it is called inaccessible.
- ✓ An empty room whose state can be defined by its temperature is an example of an accessible environment.

## 8. KNOWN VS UNKNOWN

- ✓ Known and unknown are not actually a feature of an environment, but it is an agent's state of knowledge to perform an action.
- ✓ In a known environment, the results for all actions are known to the agent. While in unknown environment, agent needs to learn or explore how it works in order to perform an action.

**Example:** Route planning in a GPS navigation system – Outcomes of paths are already known.

Exploring an unfamiliar maze – The agent must discover paths and outcomes while navigating.

| Task Environment          | Observable | Agents | Deterministic | Episodic   | Static  | Discrete   |
|---------------------------|------------|--------|---------------|------------|---------|------------|
| Crossword puzzle          | Fully      | Single | Deterministic | Sequential | Static  | Discrete   |
| Chess with a clock        | Fully      | Multi  | Deterministic | Sequential | Semi    | Discrete   |
| Poker                     | Partially  | Multi  | Stochastic    | Sequential | Static  | Discrete   |
| Backgammon                | Fully      | Multi  | Stochastic    | Sequential | Static  | Discrete   |
| Taxi driving              | Partially  | Multi  | Stochastic    | Sequential | Dynamic | Continuous |
| Medical diagnosis         | Partially  | Single | Stochastic    | Sequential | Dynamic | Continuous |
| Image analysis            | Fully      | Single | Deterministic | Episodic   | Semi    | Continuous |
| Part-picking robot        | Partially  | Single | Stochastic    | Episodic   | Dynamic | Continuous |
| Refinery controller       | Partially  | Single | Stochastic    | Sequential | Dynamic | Continuous |
| Interactive English tutor | Partially  | Multi  | Stochastic    | Sequential | Dynamic | Discrete   |

Examples of task environments and their characteristics.



# **THE STRUCTURE OF AGENTS**

# THE STRUCTURE OF AGENTS

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

$$\text{Agent} = \text{Architecture} + \text{Agent program}$$

**Architecture:** Architecture is machinery that an AI agent executes on.

**Agent Function:** Agent function is used to map a percept to an action.

$$f:P^* \rightarrow A$$

**Agent program:** Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function  $f$ .

## TYPES OF AI AGENTS

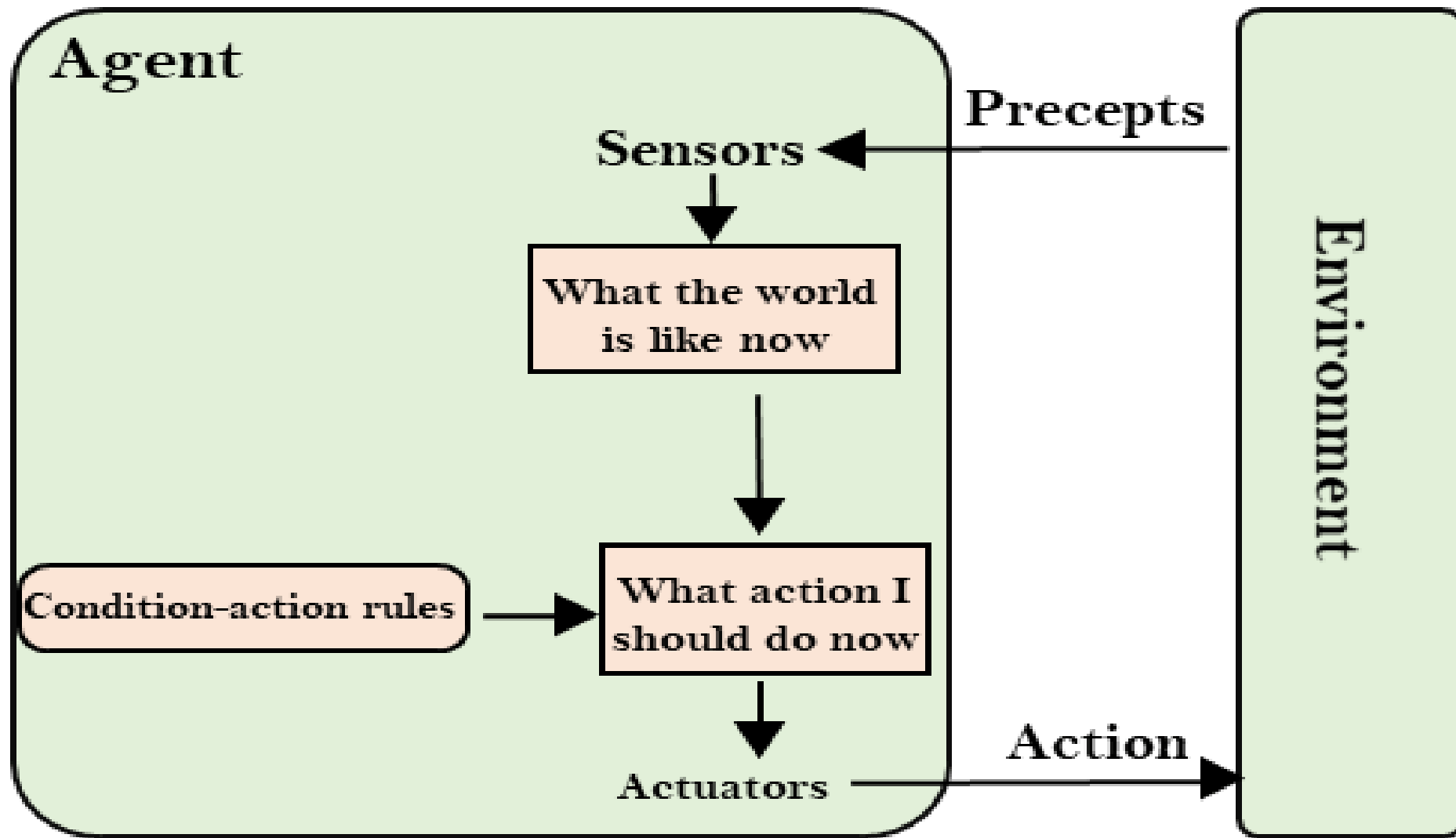
Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time.

These are given below:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

## 1. SIMPLE REFLEX AGENT:

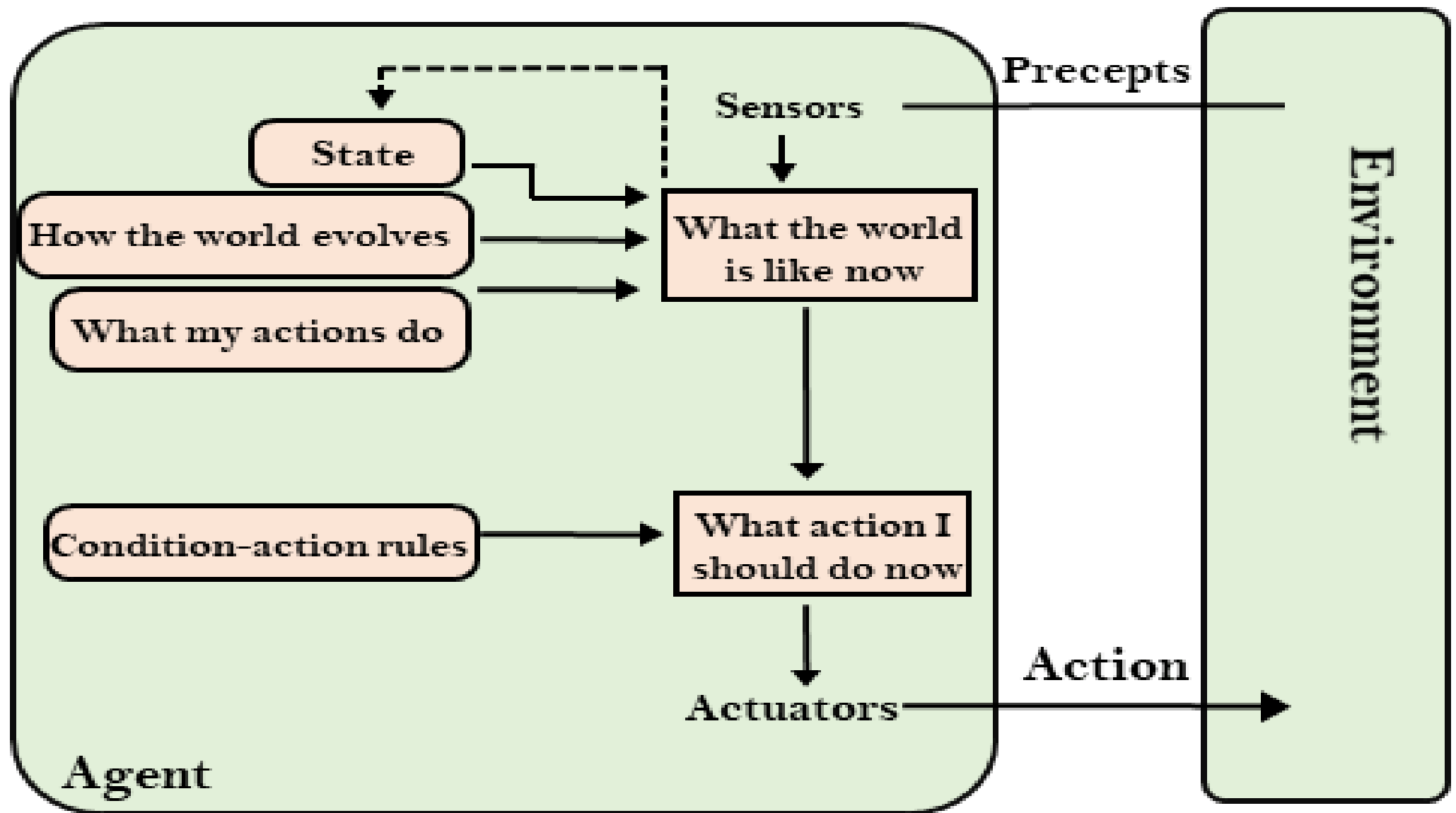
- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history.
- These agents only succeed in the fully observable environment.
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.
- Problems for the simple reflex agent design approach:
  - ✓ They have very limited intelligence
  - ✓ They do not have knowledge of non-perceptual parts of the current state
  - ✓ Mostly too big to generate and to store.
  - ✓ Not adaptive to changes in the environment.



**Schematic diagram of a simple reflex agent.**

## 2. MODEL-BASED REFLEX AGENT

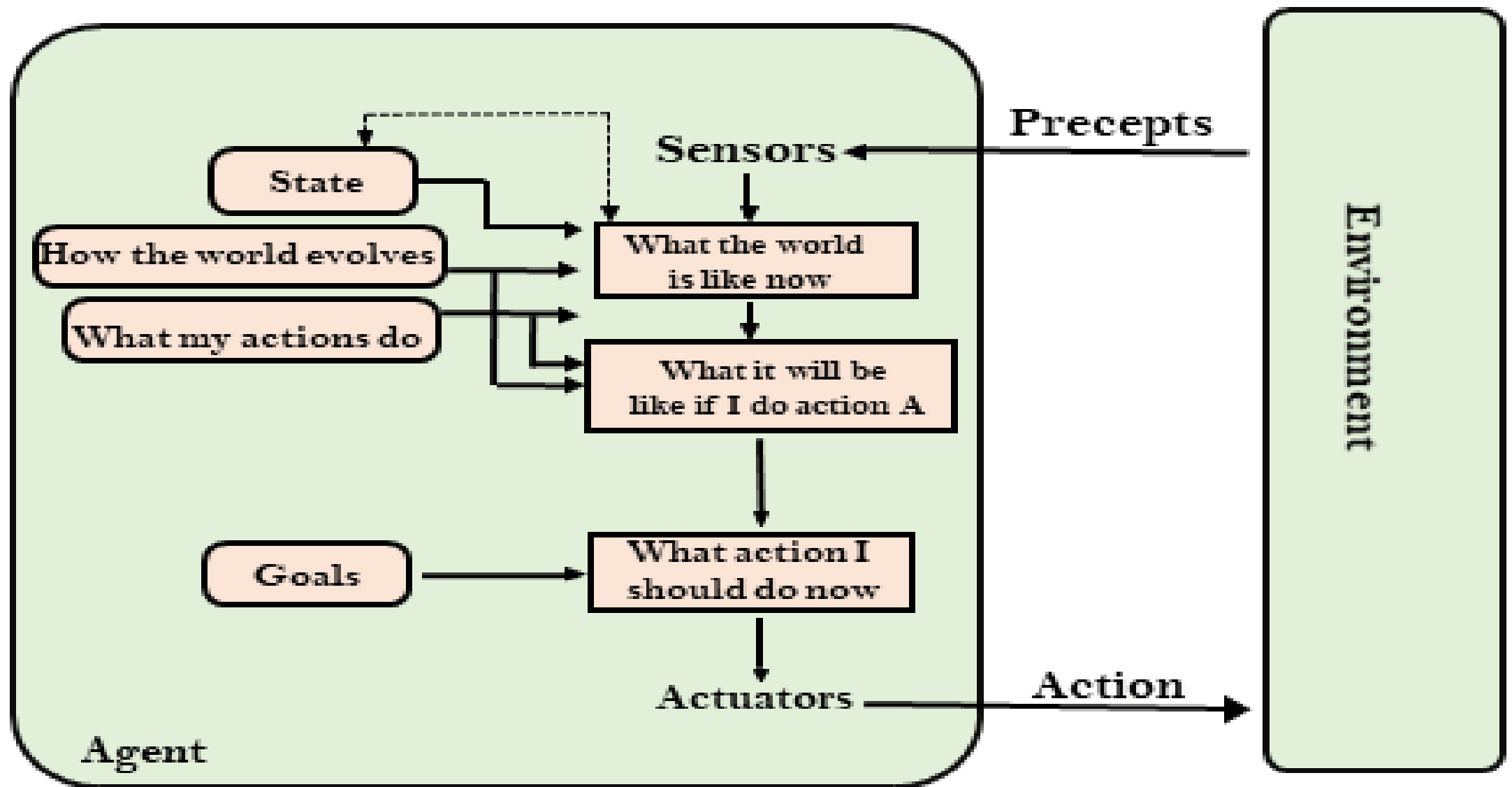
- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
  - ✓ **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  - ✓ **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
  - ✓ How the world evolves
  - ✓ How the agent's action affects the world.



**A model-based reflex agent.**

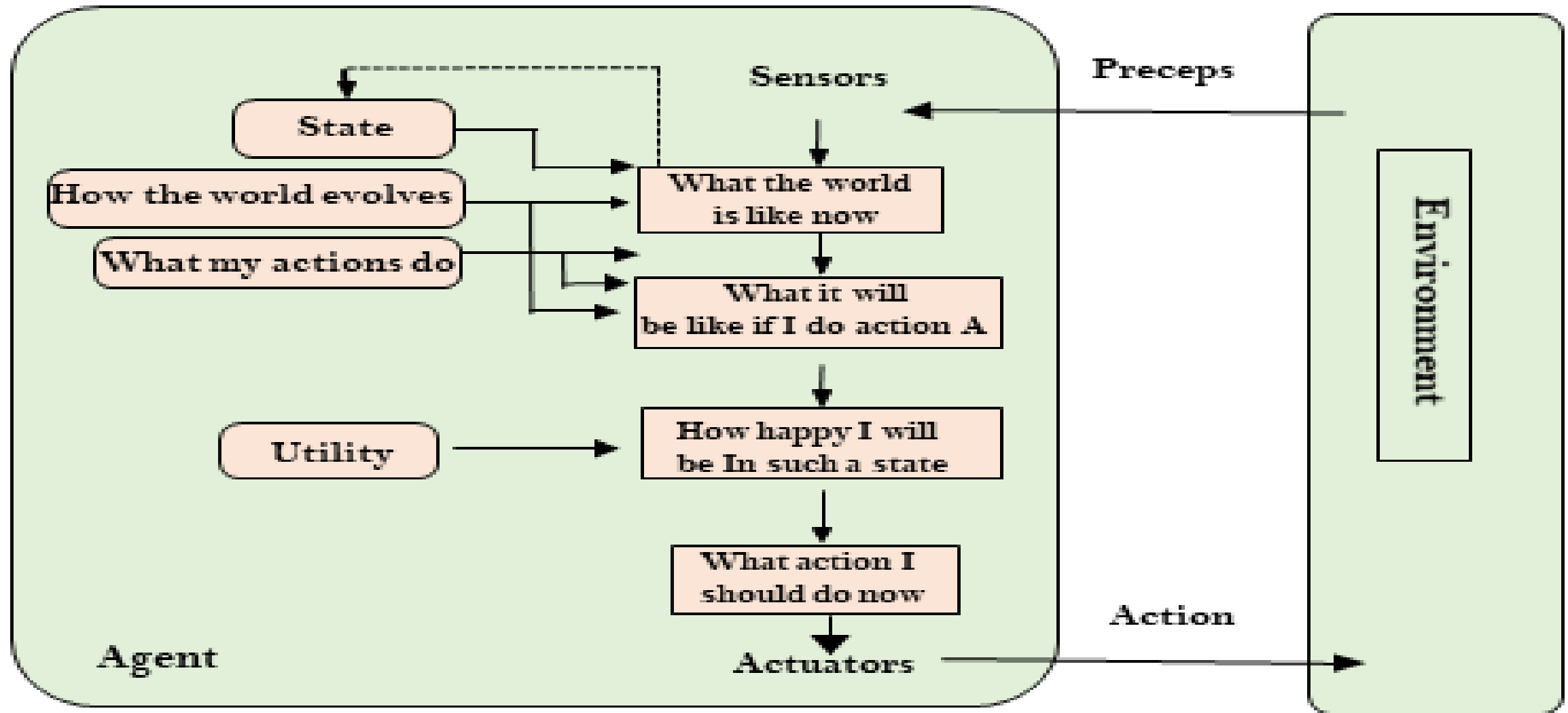
### 3. GOAL-BASED AGENTS

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.



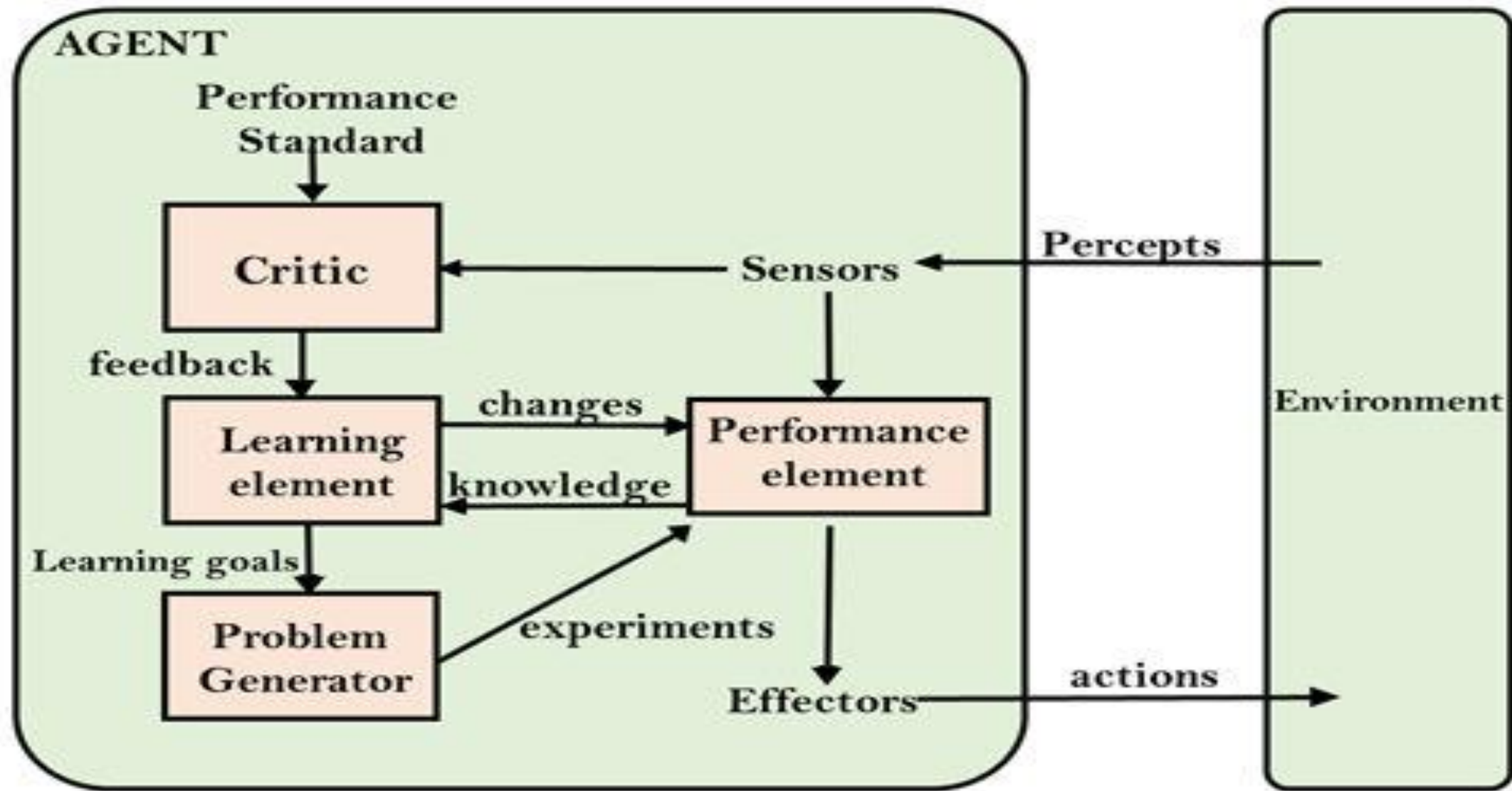
## 4. UTILITY-BASED AGENTS

- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.



## 5. LEARNING AGENTS

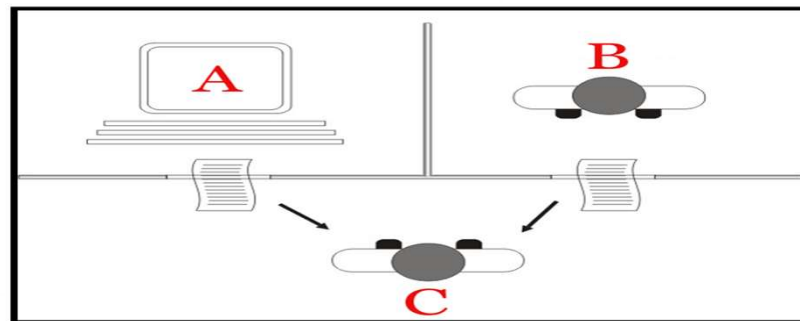
- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
  - ✓ **Learning element:** It is responsible for making improvements by learning from environment
  - ✓ **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  - ✓ **Performance element:** It is responsible for selecting external action
  - ✓ **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.



**A general learning agent.**

# Turing Test in AI

- The **Turing test**, developed by Alan Turing in 1950, is a test of a **machine's ability to exhibit intelligent behavior equivalent** to, or indistinguishable from, **that of a human**.
- This test was invented to check **“Can a Computer talk like a Humann”** ? So, he invented Turing Test as a method of inquiry in Artificial Intelligence(AI) for determining whether a computer is capable of thinking like a human being.



Now the Question is...

## **What is Turing Test?**

- Turing Test requires three terminals, each of which is physically separated from the other two. One terminal is operated by a computer, while the other two are operated by humans.
- During the test, one of the humans' functions as the questioner, while the second human and the computer function as respondents.
- The questioner interrogates the respondents within a specific subject area, using a specified format and context.
- After a preset length of time or number of questions, the questioner is then asked to decide which respondent was human and which was a computer.